PROFESSIONAL INSTALLATION,
OPERATING AND SERVICE MANUAL

SOFTENER + IRON FILTRATION

- ☐ BPFES100J
- ☐ BPFES150
- ☐ BPFES200
- ☐ BPFES100JBF
- ☐ BPFES150BF
- ☐ BPFES200BF

SOFTENER

- ☐ BP75J
- ☐ BP100J
- ☐ BP150J
- ☐ BP200
- ☐ BP250
- ☐ BP75JBF
- ☐ BP100JBF
- ☐ BP150JBF
- ☐ BP200BF
- ☐ BP250BF



PLATINUM SERIES

ELECTRONIC WATER SOFTENER
PROGRAMMED FOR PRE-FILL BRINING OPTIONS

V. 10/22/2026

Antimicrobial Technology

BRITA|PRO

JOB SPECIFICATIONS

MODEL NO. _____

INSTALLATION DATE _____

SERIAL NUMBER _____

INSTALLER NAME _____ PHONE _____

ADDRESS _____

WATER TEST AT TIME OF INSTALLATION

_____ Incoming PSI

_____ Hardness $CaCo_3$ (gpg) Other: _____

_____ Iron (ppm) _____

_____ pH _____

SIZING INFORMATION

All Water is Softened Except:

_____ Rear Hose Bib _____ Front Hose Bib _____ Kitchen Cold _____ Toilets _____ All Cold
_____ Other _____

The average family uses 50 gallons per person daily for all water uses in the home.

_____ Daily Water Usage (Gallons/Person)

x _____ Family Size (Number of people in family)

= _____ Total Gallons Per Day

x _____ Grains Per Gallon of hardness

(Note: Add 4 grains per gallon of hardness for each ppm iron for total compensated hardness)

= _____ Total Grains per Day

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GENERAL INSTALLATION AND SERVICE WARNINGS

The control valve, fittings and/or bypass are designed to accommodate minor plumbing misalignments but are not designed to support the weight of a system or the plumbing.

HYDROCARBONS SUCH AS KEROSENE, BENZENE, GASOLINE, ETC., MAY DAMAGE PRODUCTS THAT CONTAIN O-RINGS OR PLASTIC COMPONENTS. EXPOSURE TO SUCH HYDROCARBONS MAY CAUSE THE PRODUCTS TO LEAK. DO NOT USE THE PRODUCT(S) CONTAINED IN THIS DOCUMENT ON WATER SUPPLIES THAT CONTAIN HYDROCARBONS SUCH AS KEROSENE, BENZENE, GASOLINE, ETC.

THIS WATER METER SHOULD NOT BE USED AS THE PRIMARY MONITORING DEVICE FOR CRITICAL OR HEALTH EFFECT APPLICATIONS

Do not use Vaseline, oils, other hydrocarbon lubricants or spray silicone anywhere. A silicone lubricant may be used on black o-rings.

The nuts and caps are designed to be unscrewed or tightened by hand or with the clack service wrench.
Do not use a pipe wrench to tighten or loosen nuts or caps. **Do not** place a screwdriver in the slots on caps and/or tap with a hammer.

Do not use pipe dope or other sealants on threads. Use Teflon tape on the threaded inlet, outlet and drain fittings. Teflon tape is not necessary on the nut connection or caps because of o-ring seals.

After completing any valve maintenance involving the drive assembly or the drive cap assembly and pistons unplug power source jack from the printed circuit board (black wire) and plug back in or press and hold NEXT and REGEN buttons for 3 seconds.

This resets the electronics and establishes the service piston position. The display should flash all wording, then flash the software version and then reset the valve to the service position.

All plumbing should be done in accordance with local plumbing codes. The pipe size for the drain line should be a minimum of ½". Backwash flow rates in excess of 7 gpm or length in excess of 20' require ¾" drain line.

Solder joints near the drain must be done prior to connecting the drain line flow control fitting. Leave at least 6" between the drain line control fitting and solder joints when soldering pipes that are connected on the drain line control fitting. Failure to do this could cause interior damage to the drain line flow control fitting.

When assembling the installation fitting package (inlet and outlet), connect the fitting to the plumbing system first and then attach the nut, split ring and o-ring. Heat from soldering or solvent cements may damage the nut, split ring or o-ring. Solder joints should be cool and solvent cements should be set before installing the nut, split ring and o-ring. Avoid getting primer and solvent cement on any part of the o-rings, split rings, bypass valve or control valve.

Plug into an electrical outlet. Note: All electrical connections must be connected according to local codes. (Be certain the outlet is uninterrupted.)

Install grounding strap on metal pipes.

This glass filled Noryl (or equivalent) fully automatic control valve is designed as the primary control center to direct and regulate all cycles of a water softener or filter. The control valve can be set to regenerate on demand (consumption of a predetermined amount of water) and/or as a time clock (passage of a particular number of days). The control valve can be set so that a softener can meet the Water Quality Association (WQA) Standard S100 or NSF/ANSI Standard 44 efficiency rating.

It is not recommended to change control valves from downflow to upflow brining or vice versa in the field. The valve bodies for downflow and upflow are unique to the regeneration type and should not be interchanged. A mismatch of valve body and regeneration piston will result in hard water bypass during service.

The control valve is compatible with a variety of regenerants and resin cleaners. The control valve is capable of routing the flow of water in the necessary paths to regenerate or backwash water treatment systems. The injector regulates the flow of brine or other regenerants. The control valve regulates the flow rates for backwashing, rinsing, and the replenishing of treated water into a regenerant tank, when applicable. The control valve uses no traditional fasteners (e.g. screws); instead clips, threaded caps and nuts and snap type latches are used. Caps and nuts only need to be firmly hand tightened because radial seals are used. Tools required to service the valve include one small blade screwdriver, one large blade screwdriver, pliers and a pair of hands. A plastic wrench is available which eliminates the need for screwdrivers and pliers. Disassembly for servicing takes much less time than comparable products currently on the market. Control valve installation is made easy because the distributor tube can be cut ½" above to ½" below the top of tank thread. The distributor tube is held in place by an o-ring seal and the control valve also has a bayonet lock feature for upper distributor baskets.

The AC adapter comes with a 15 foot power cord and is designed for use with the control valve. The AC adapter is for dry location use only. The control valve remembers all settings until the battery power is depleted if the power goes out. After the battery power is depleted, the only item that needs to be reset is the time of day; other values are permanently stored in the nonvolatile memory. The control valve battery is not rechargeable but is replaceable.

Allow one foot of clearance to service the valve.

WATER QUALITY: If sand or sediment is present in the water supply, a sediment filter should be installed ahead of the water softener.

PRE-INSTALLATION REVIEW

Your water softener has been designed to adequately reduce hardness from levels up to 100 grains per gallon. Ferrous bicarbonate iron levels up to 0.5 ppm can also be reduced. This is iron that is dissolved in water and not visible to the eye in a freshly drawn sample. After standing in contact with air, the ferrous iron will become oxidized to the ferric state and start to precipitate as a reddish brown floc. It can be seen and may cause discolored water. Air must not come in contact with water until after it has passed through the water softener. In some cases, additional treatment equipment may be needed to treat water having special characteristics, such as: ferric hydroxide iron, iron bacteria, low pH, tastes and odors, etc. Consult your dealer if you have any questions. **This water softener is not to be used for treating water that is microbiologically unsafe or of unknown quality without adequate disinfection before or after treatment.**

WATER PRESSURE: A minimum of **30** pounds of water pressure (psi) is required for regeneration. Maximum **80** psi. CAUTION: the softener cannot be subject to a vacuum due to loss of pressure (such as a water main break or submersible well pump check valve failure).

WATER TEMPERATURE: The range of water temperature is 40°F to 100°F. DO NOT install any water softener with less than 10 feet of piping between its outlet and the inlet of a water heater.

AMBIENT TEMPERATURE: DO NOT locate softener where it or its connections (including the drain and overflow lines) will ever be subject to room temperatures under 33°F.

ELECTRICITY: An uninterrupted 120 volt 60Hz source is required. *Make sure electrical source is not on a timer or switch.* All electrical connections must be connected according to local codes. The plug-in transformer is for dry locations only. Surge protection is recommended with all electrical connections.

DRAIN: All plumbing should be done in accordance with local plumbing codes. The distance between the drain and the water softener should be as short as possible. The pipe size for the drain line should be a minimum of 1/2” (inside diameter of pipe).

SOFTENING: It is recommended that the softener be installed to soften both the hot and cold water supply. A separate hard water faucet may be plumbed for drinking purposes if desired. Outside faucets should be left on hard water.

BYPASS: A bypass valve should be installed so that water will be available if it should be necessary to shut off the pressure in order to service the softener.

BYPASS VALVE

The bypass valve is typically used to isolate the control valve from the plumbing system's water pressure in order to perform control valve repairs or maintenance. The bypass valve is particularly unique in the water treatment industry due to its versatility and state of the art design features. The 1" full flow bypass valve incorporates four positions, including a diagnostic position that allows service personnel to work on a pressurized system while still providing untreated bypassed water to the facility or residence. Its completely non-metallic, all-plastic design allows for easy access and serviceability without the need for tools.

The bypass body and rotors are glass filled Noryl® (or equivalent) and the nuts and caps are glass filled polypropylene. All seals are self-lubricating EPDM to help prevent valve seizing after long periods of non-use. Internal o-rings can easily be replaced if service is required.

The bypass consists of two interchangeable plug valves that are operated independently by red arrow-shaped handles. The handles identify the flow direction of the water. The plug valves enable the bypass valve to operate in four positions.

OPERATION:

1. Service Operation Position: The inlet and outlet handles point in the direction of flow indicated by the engraved arrows on the control valve. Water flows through the control valve during normal operation and this position also allows the control valve to isolate the media bed during the regeneration cycle. *(see figure 1)*

2. Bypass Position: The inlet and outlet handles point to the center of the bypass, the control valve is isolated from the water pressure contained in the plumbing system. Untreated water is supplied to the plumbing system. *(see figure 2)*

3. Diagnostic Position: The inlet handle points in the direction of flow and the outlet handle points to the center of bypass valve, system water pressure is allowed to the control valve and the plumbing system while not allowing water to exit from the control valve to the plumbing. *(see figure 3)*

4. Shut Off Position: The inlet handle points to the center of the bypass valve and the outlet handle points in the direction of flow, the water is shut off to the plumbing system. If water is available on the outlet side of the softener it is an indication of water bypass around the system (i.e. a plumbing connection somewhere in the building bypasses the system). *(see figure 4)*

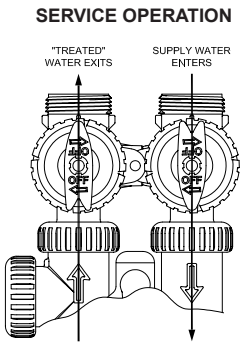


figure 1

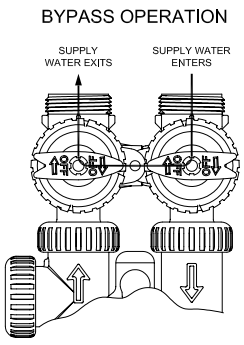


figure 2

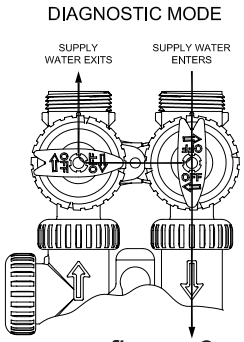


figure 3

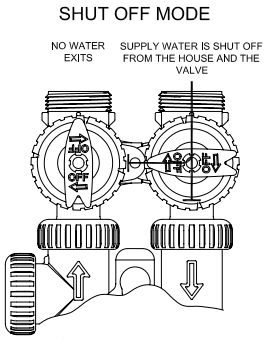


figure 4

The working parts of the bypass valve are the rotor assemblies that are contained under the bypass valve caps. Before working on the rotors, make sure the system is depressurized. Turn the red arrow shaped handles towards the center of the bypass valve and back several times to ensure rotor is turning freely.

The nuts and caps are designed to be unscrewed or tightened by hand. If necessary, pliers can only be used to unscrew the nut or cap. Do not use a pipe wrench to tighten or loosen nuts or caps. Do not place screwdriver in slots on caps and/or tap with a hammer. To access the rotor, unscrew the cap and lift the cap, rotor and handle out as one unit. Twisting the unit as you pull it out will help to remove it more easily. There are three o-rings: one under the rotor cap, one on the rotor stem and the rotor seal. Replace worn o-rings. Clean rotor. Reinstall rotor.

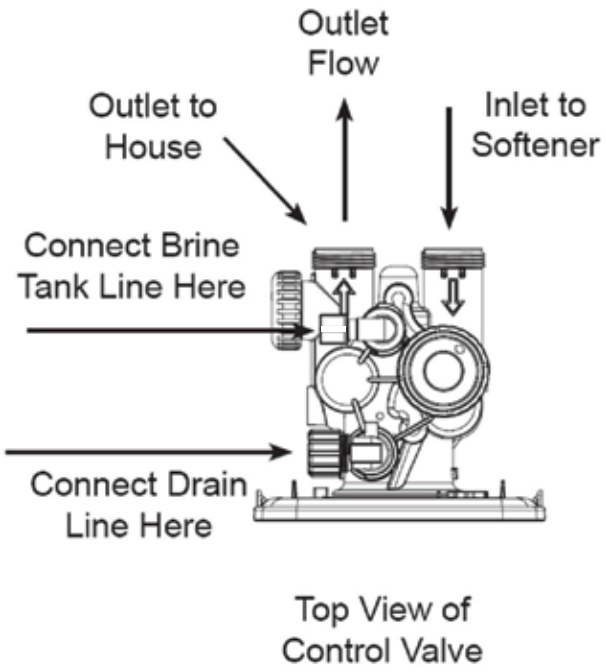
When reinstalling the red arrow handles be sure that:

1. The handle pointers are lined up with the control valve body arrows, and the rotor seal o-ring and retainer on both rotors face to the right when being viewed from the front of the control valve; or
2. Arrows point toward each other in the bypass position.

Since the handles can be pulled off, they could be accidentally reinstalled 180° from their correct orientation. To install the red arrow handles correctly, keep the handles pointed in the same direction as the arrows engraved on the control valve body while tightening the bypass valve caps.

INSTALLATION INSTRUCTIONS

(All electrical & plumbing should be done in accordance to all local codes)

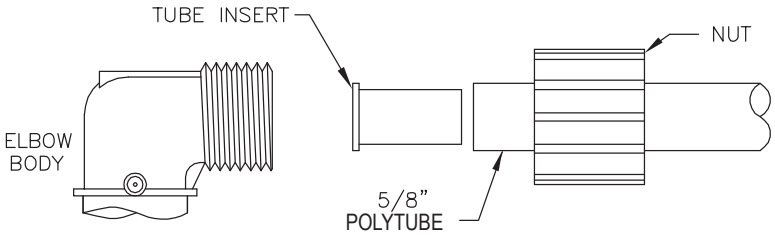


1. Place the softener where you want to install it, making sure it is on a clean, level and firm base.

2. Do all necessary plumbing (inlet to inlet, outlet to outlet, and drain line to drain). The control valve, fittings and/or bypass are designed to accommodate minor plumbing misalignments but are not designed to support the weight of a system or the plumbing.

3. When assembling the installation fitting package (inlet and outlet), connect the fitting to the plumbing system first and then attach the nut, split ring and o-ring. Heat from soldering or solvent cements may damage the nut, split ring or o-ring. Solder joint should be cool and solvent cements should be set before installing the nut, split ring and o-ring. Avoid getting primer and solvent cement on any part of the o-rings, split rings, bypass valve or control valve.

4. A jumper ground wire should be installed between the inlet and outlet pipe whenever the metallic continuity of a water distribution piping system is interrupted. Install grounding strap on metal pipes.

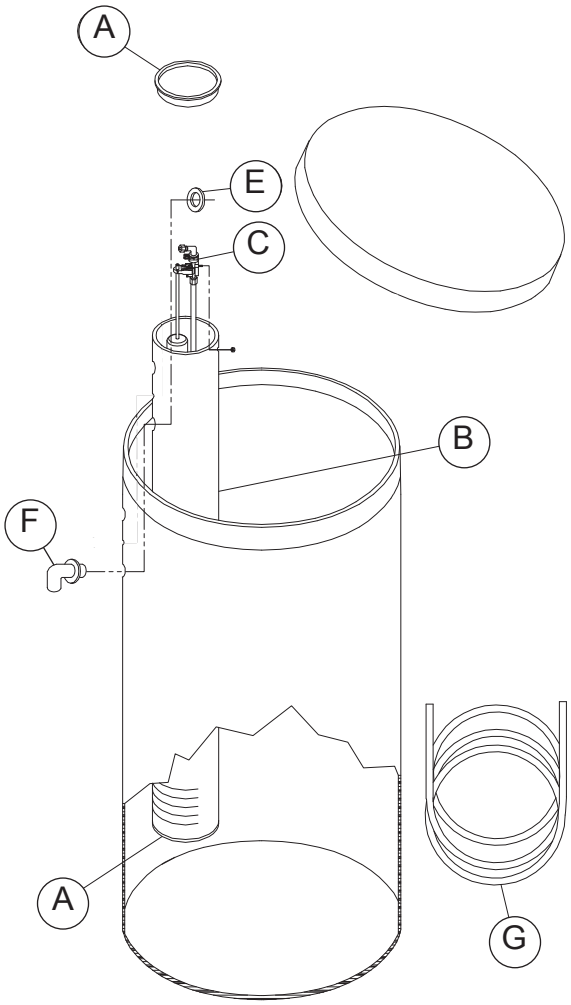


DRAIN LINE FITTING CONNECTION USING 5/8" POLY TUBE

5. The drain connection may be made using either 5/8” polytube (see below) or a 3/4” female adapter. The polytube insert is shipped attached to the drain line elbow’s locking clip. Press the insert into the drain line tubing (tubing not provided). Loosen the nut of the drain line elbow. Press the 5/8” polytube with insert into the drain line elbow until it seats on the back of the fitting. Tighten the nut. If soldering, joints near the drain must be done prior to connecting the drain line flow control fitting. Leave at least 6” between the drain line control fitting and solder joints when soldering pipes that are connected on the drain line control fitting. Failure to do this could cause interior damage to the drain line flow control fitting. Never insert a drain line into a drain, sewer line, or trap. Always allow an air gap between the drain line and the wastewater to prevent the possibility of sewage being back-siphoned into the softener.

INSTALLATION INSTRUCTIONS CONT'D

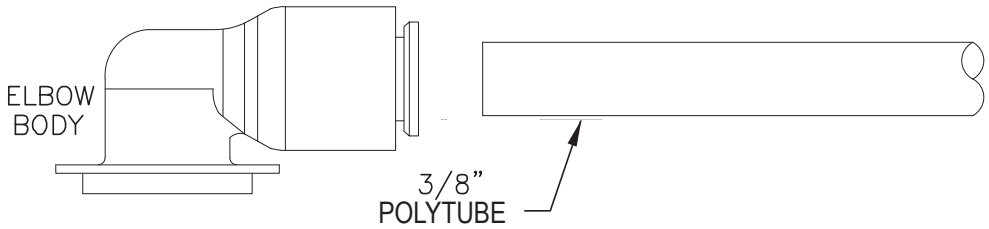
(All electrical & plumbing should be done in accordance to all local codes)



6. Assemble the brine tank. Install a brine well cap (Item A) on the bottom of the brine well (Item B). Remove the brine valve assembly (Item C) from the brine well by removing the nut (Item D). Attach the brine well to the brine tank with the two-piece overflow (Items E & F) using the lower holes in both pieces, as shown. Reinstall the brine valve assembly. Push brine line polytube (Item G) through the upper holes in the brine tank and brine well and press the polytube into the elbow fitting.

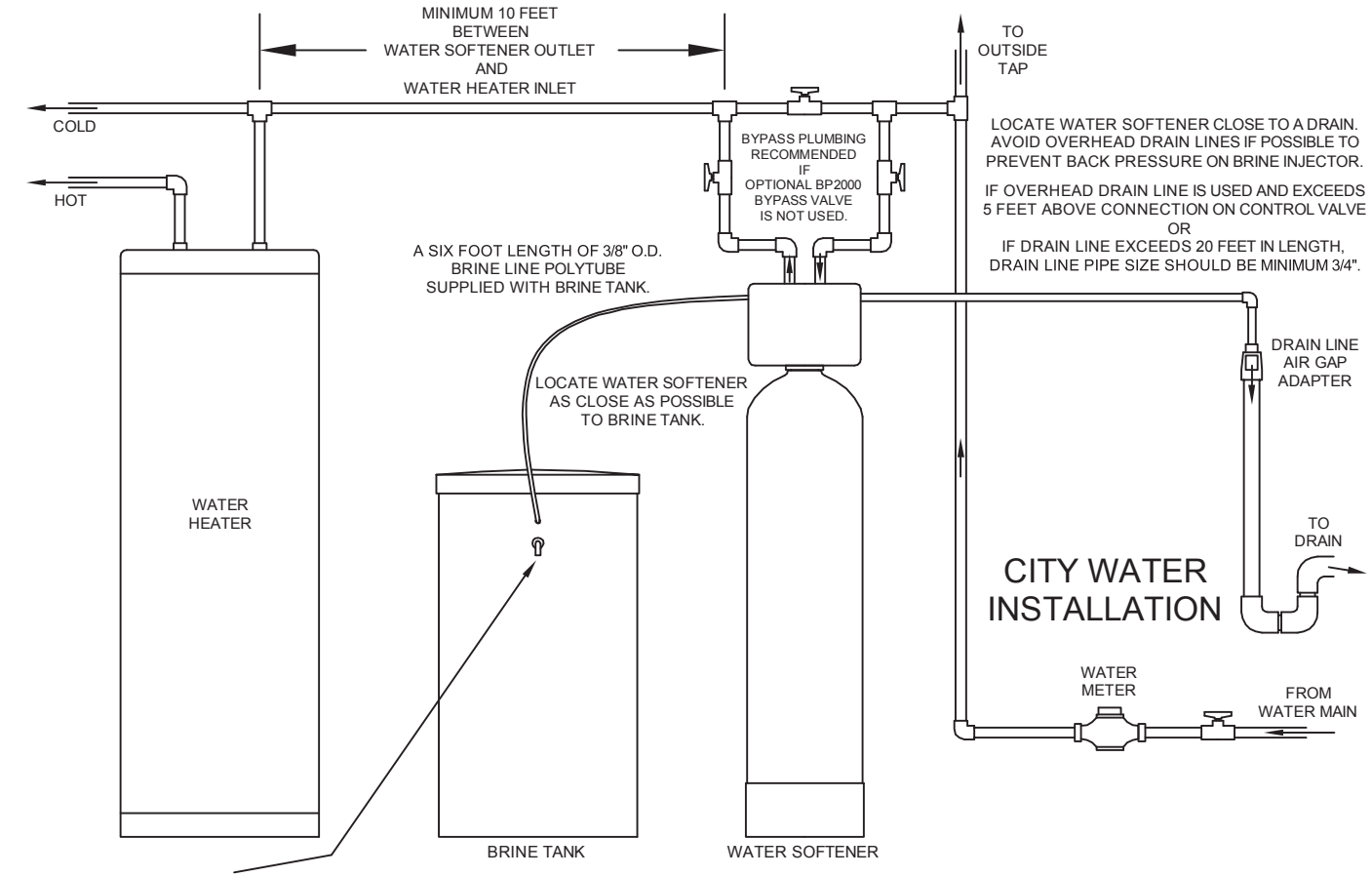
7. The brine refill flow control assembly is installed in an easy to access refill elbow located on top of the control valve. The refill flow control assembly is attached to the control valve with a locking clip. The locking clip allows the elbow to rotate 270 degrees so the outlet can be oriented towards the brine tank.

8. A 1/2" (inside diameter, not provided) gravity drain line should be connected to the overflow fitting on the side of the brine tank. This overflow is in case of a malfunction in the brine shut off. If the unit is installed where water may flow in the event of an overflow and cause water damage, connect a length of flexible tubing and run to a drain below the level of the overflow. **(Do not connect the tubing to the drain line on the control valve. Do not run tubing above overflow height at any point.)**

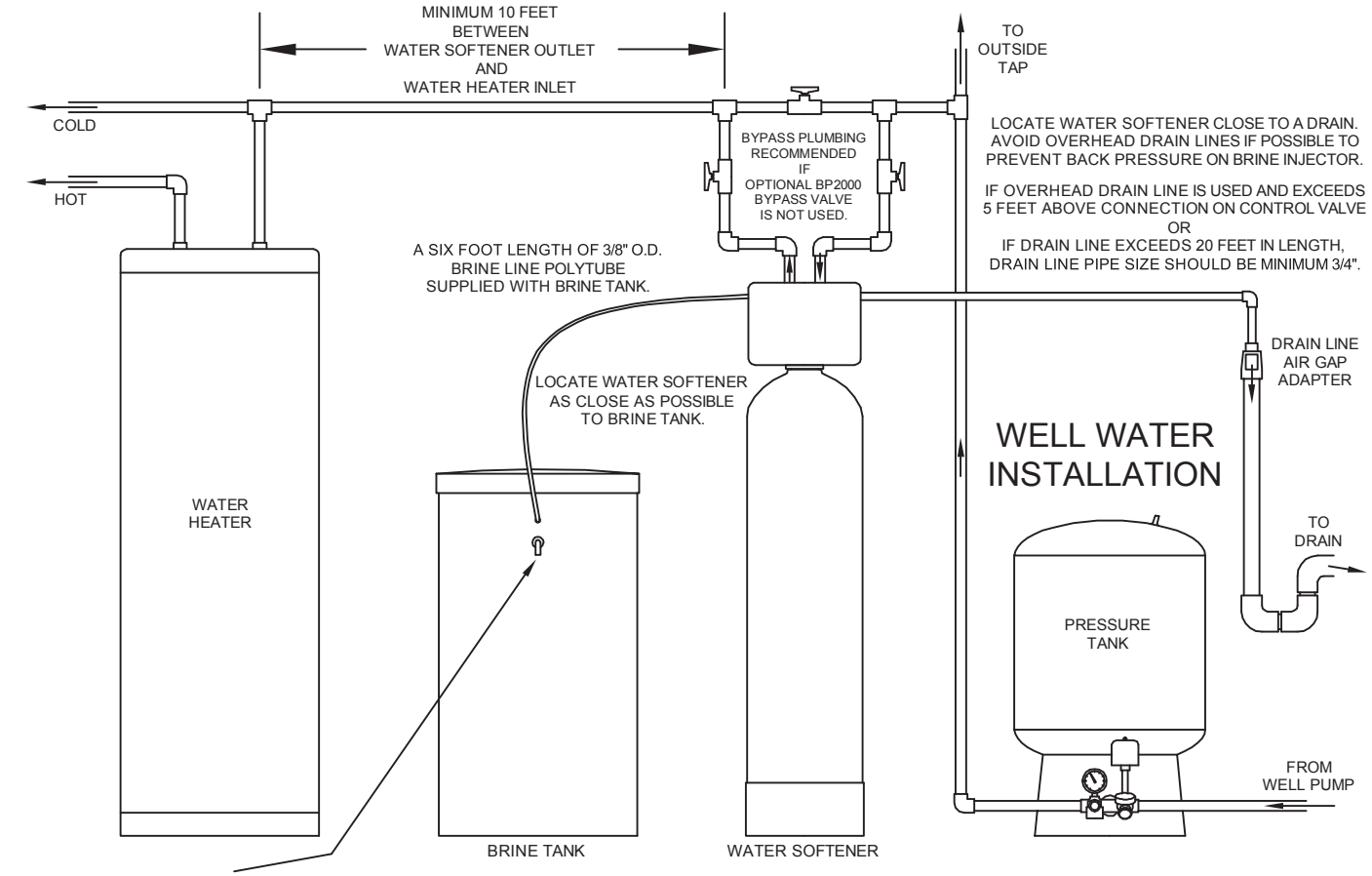


BRINE LINE FITTING CONNECTIONS

9. Connect the brine line polytubing found with the brine tank to the brine connection on the control valve. The control valve has a standard refill elbow to which a 3/8" flexible tube can be connected, see below. Press the polytube into the brine elbow fitting. Make sure the floor is clean beneath the brine tank and that it is level and smooth.



OVERFLOW GRAVITY DRAIN - ONLY USED IN CASE OF MALFUNCTION IN THE BRINE SHUTOFF. DO NOT CONNECT TO CONTROL VALVE DRAIN LINE.
IF UNIT IS INSTALLED WHERE OVERFLOW COULD CAUSE WATER DAMAGE, CONNECT TUBING AND RUN TO FLOOR DRAIN. DO NOT RUN TUBING ABOVE OVERFLOW HEIGHT.



OVERFLOW GRAVITY DRAIN - ONLY USED IN CASE OF MALFUNCTION IN THE BRINE SHUTOFF. DO NOT CONNECT TO CONTROL VALVE DRAIN LINE.
IF UNIT IS INSTALLED WHERE OVERFLOW COULD CAUSE WATER DAMAGE, CONNECT TUBING AND RUN TO FLOOR DRAIN. DO NOT RUN TUBING ABOVE OVERFLOW HEIGHT.

PLACING SOFTENER INTO SERVICE

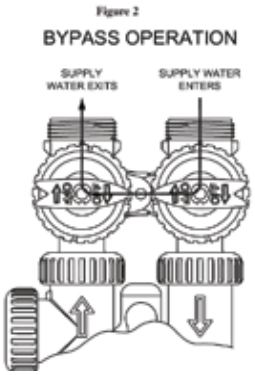
WATER SOFTENER & MID-PLATE STARTUP PROCEDURE

STEP 1

Before Startup

1.) Verify incoming water pressure does not exceed 80 psi.
2.) Verify the inlet and outlet connections are correct and secure.
3.) Verify the drain line is connected, secured, and unobstructed.
4.) Verify the brine line is properly connected.
5.) Verify the brine tank overflow connection is installed where required.
6.) Confirm the system is in Bypass.
7.) Verify electrical power is available.

→ **IMPORTANT:** Inspect plumbing, drain, and brine connections for leaks throughout startup.



BYPASS OPERATION
both handles are positioned toward the center of the bypass.

STEP 2

Restore Water & Flush

1.) Open a downstream cold-water faucet.
2.) Slowly open the building's main water supply.
3.) Flush until the water runs clear and all air is expelled.
4.) Close the faucet.

STEP 3

Power & Verify Valve

1.) Plug the system into an electrical outlet.
2.) Pull the battery tab located behind the faceplate.
3.) Verify the valve is properly programmed.
4.) Verify the correct time of day.

⚠

CAUTION: Do not remove the battery tab before plugging the valve into an electrical outlet. Valve programming may be damaged if the battery tab is removed before the valve is plugged into an electrical outlet.

STEP 4

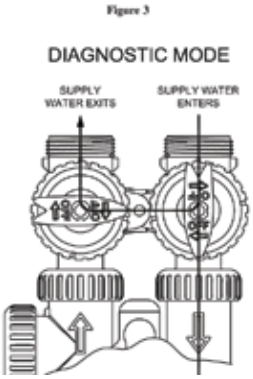
Initiate Manual Regeneration

1.) Press and hold REGEN to initiate regeneration.
2.) Allow the valve motor to stop at each cycle position.
3.) Press REGEN after each motor stop to advance through the regeneration sequence.
4.) Continue until the valve reaches the second Backwash cycle.

STEP 5

Fill & Purge Mineral Tank

1.) At the second Backwash, unplug the valve to hold it in Backwash.
2.) Crack the inlet side of the bypass valve approximately 1/8 turn.
3.) Allow the mineral tank to fill slowly.
4.) As the air clears, gradually open the inlet farther.
5.) Continue until the inlet is fully open and full backwash flow is established.
6.) Verify steady flow to drain.
7.) Purge trapped air from the mineral tank by briefly closing the inlet valve. Leave the inlet valve closed for at least 1 minute, then slowly reopen it to the Normal Operation position. Repeat as necessary until air is expelled.
8.) Continue backwashing until the discharge is clear and free of air.
9.) Inspect the system and drain connections for leaks.



DIAGNOSTIC MODE
Inlet open and outlet closed. During initial filling, begin with the inlet approximately 1/8 turn open and gradually open it fully.

STEP 6

Complete Rinse

1.) Plug the control valve back into power.
2.) Press REGEN to advance the valve to Rinse.
3.) Allow the complete 4-minute Rinse cycle to finish.

STEP 7

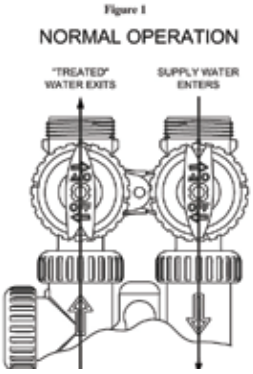
Add Salt → **IMPORTANT:** Use solar salt only.

1.) Add high-quality solar salt to the brine tank, filling it approximately 1/2 to 2/3 full.
2.) Verify that salt is not interfering with the brine well, safety float, or brine line.
3.) Replace the brine tank lid.

STEP 8

System Verification

1.) Verify the control valve is in Normal Operation and displays the correct time of day.
2.) Inspect the system and all plumbing connections for leaks.
3.) Confirm the installation area is clean and the system is ready for normal operation.
4.) Test the treated water and verify the system is producing soft water.
5.) Operate all plumbing fixtures to verify proper water flow and system operation.
6.) Review system operation and basic maintenance with the customer, answer any questions, and turn the system over to the customer.



NORMAL OPERATION
Inlet and outlet handles are positioned in the direction of flow.

STEP 9

NOTE: Use only plain, unscented household chlorine bleach. Do Not use scented, splashless, gel, or bleach containing additional cleaning additives.

Sanitize System

1.) Add 1.5 fl oz (45 mL) of plain, unscented 5.25% household chlorine bleach per cubic foot of resin to the brine well.
2.) Press REGEN button 1 time. This sets valve to regenerate at 2:00 AM overnight.

GENERAL OPERATION

Note: As an energy saving feature, the control will automatically turn off all SOLID BLUE or SOLID GREEN display illumination and keypad illumination after about 5 minutes of the last keypad button push. Any further keypad touch will cause the re-illumination of the display and keypad, and re-activate keypad control.

User Displays

When the system is in normal service mode, one of up to five available User Displays will be shown. Pressing NEXT will alternate between the following displays:

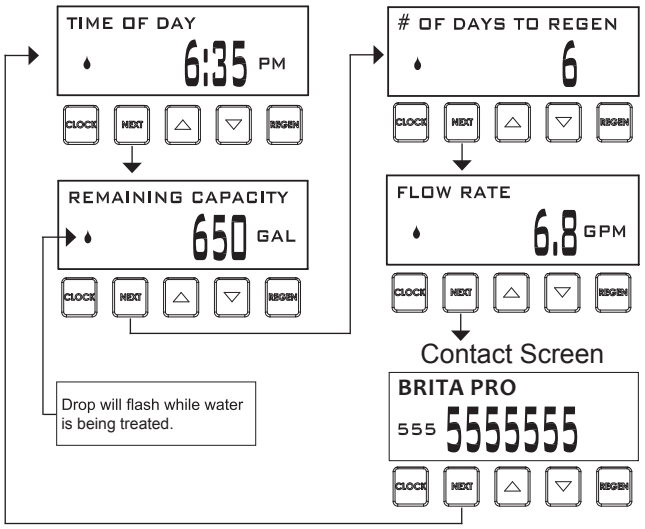
- Current time of day
- Treated water flow rate
- Service contact name and phone number (if entered)
- Remaining Capacity of treated water available
- Remaining days to regeneration (if Day Override is programmed)

Pressing the ▼ button while in the Capacity Remaining or Days Remaining displays will decrease the capacity remaining in ten gallon increments or the days remaining in one day increments.

To clear the Service Call reminder, press the ▲ and ▼ buttons simultaneously while the number and banner text screen is displayed.

If the system has called for a regeneration that will occur at the preset time of regeneration, the words REGEN TODAY will alternate with the header on the display.

Utilizing the control valve's built-in water meter, a water drop flashes on the display when water is being treated (i.e. water is flowing through the system).



SET TIME OF DAY

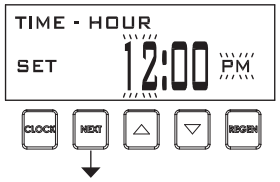
Current time of day needs to be entered during initial installation, and adjusted when daylight saving time begins or ends. If an extended power outage occurs and depletes the on-board non-rechargeable coin cell battery, when power resumes the time of day should be reset and battery replaced.

STEP 1



STEP 1 – Press CLOCK.

STEP 2



STEP 2 - Current Time (hour): Set the hour of the day using ▼ or ▲. AM/PM toggles after 12. Press NEXT to go to Step 3.

STEP 3

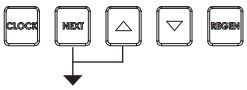


STEP 3 - Current Time (minutes): Set the minutes of the day using ▼ or ▲. Press NEXT to exit Set Time of Day. Press REGEN to return to previous step.

RETURN TO NORMAL MODE

ADJUST HARDNESS, DAYS BETWEEN REGENERATIONS, OR TIME OF REGENERATION

STEP 1



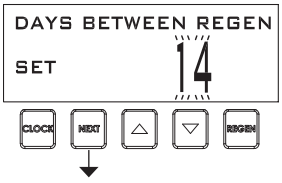
STEP 1 - Press NEXT and ▲ simultaneously for 3 seconds.

STEP 2



STEP 2 – Hardness: Set the amount of hardness in grains of hardness as calcium carbonate per gallon using ▲ or ▼ buttons. The default is 20 with value ranges from 1 to 150 in 1 grain increments. Note: The grains per gallon can be increased if soluble iron needs to be reduced. Press NEXT to go to step 3. Press REGEN to exit Installer Display Settings.

STEP 3



STEP 3 – Day Override: Set the <u>maximum</u> number of days between regenerations. If value set to “OFF”, regeneration initiation is based solely on volume used. If value is set as a number (allowable range from 1 to 28) a regeneration initiation will be called for on that day even if sufficient volume of water were not used to call for a regeneration. Set Day Override using ▲ or ▼:

- number of days between regeneration (1 to 28); or
- “OFF”.

Press NEXT to go to step 4. Press REGEN to return to previous step.

STEP 4



STEP 4 – Next Regeneration Time (hour): Set the hour of day for regeneration using ▲ or ▼ buttons. AM/PM toggles after 12. The default time is 2:00 AM. Press NEXT to go to Step 5. Press REGEN to return to previous step.

STEP 5



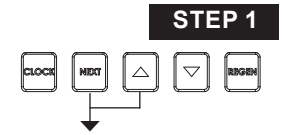
STEP 5 – Next Regeneration Time (minutes): Set the minutes of day for regeneration using ▲ or ▼ buttons. Press NEXT exit Installer Display settings. Press REGEN to return to previous step.

RETURN TO NORMAL MODE

LOW BATTERY

A non-rechargeable coin cell battery is located on the circuit board, used only to maintain the time of day during power outages (all other information will be stored in memory no matter how long the power outage). The screen displays LOW BATTERY when the battery needs to be replaced. The screen will remain illuminated solid blue when LOW BATTERY is displayed. Initially LOW BATTERY display will alternate with the User display, finally displaying only LOW BATTERY. User displays are still accessible by pressing NEXT.

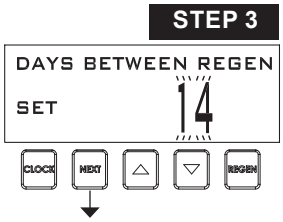
CONTACT SCREEN PROGRAMMING



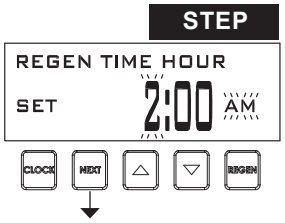
STEP 1 - Press NEXT and ▲ simultaneously for 3 seconds to access Installer Display Settings.



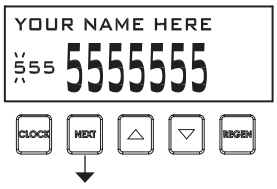
STEP 2 – Press NEXT to go to step 3.



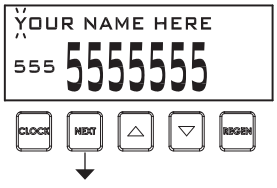
STEP 3I – Press NEXT to go to step 4.



STEP 4 – While hour is flashing, press and hold both CLOCK and ▲ to change phone number and banner text.



Phone Number - Set phone number using ▲ or ▼. Press NEXT to forward to the next digit. Press REGEN to return to previous digit.



Banner Text - Set the banner text up to a maximum of 44 characters. Use ▲ or ▼ to select letters of the alphabet, number, ampersand (&), or space in the banner text. Press NEXT to forward to the next character or to exit the Installer Display Settings.

GENERAL SPECIFICATIONS

Inlet/Outlet Fitting Options; inches	3/4 to 1.25
Drain Pipe NPT (OD Poly Tube) Size; inches	3/4 (5/8)
Working Pressure; Min to Max (PSI)	20 to 100
Water Operating Temp; Min to Max (°F)	35 to 100
AC Adaptor Input; Voltage - Hertz	120V AC - 60 Hz
AC Adaptor Output; Voltage - Current	12V AC - 500 mA
3 Volt Lithium Coin Cell Battery; type	2032
PC Board Relay Terminal Block DC Output; Voltage	12V DC **

** Relay Specifications: 12V DC Relay with a coil resistance not less than 80 ohms. If mounting the relay under the control valve cover, check for proper mounting location dimensions on the backplate.

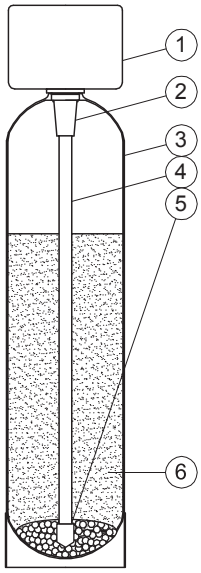


General Notes for estimating only.

SPECIFICATIONS

	CABINET SOFTENERS			SOFTENERS						IRON SOFTENERS		
MODEL NUMBER	BPC50 (PACKED BED)		BPC75	BPC100 (PACKED BED)	BP75J	BP100J	BP150J	BP200	BP250	BP300	BPFES100J	BPFES150	BPFES200
Service Flow (GPM) at 15 PSI Pressure Drop	12		16	15	11	13	14	16	16	17	13	16	17
Resin Tank Size; Diameter x Height (inches)	9 x 18		10 x 35	10 x 35	8 x 44	10 x 44	10 x 54	12 x 52	13 x 54	14 x 65	10 x 54	12 x 52	13 x 54
Resin; Cu. Ft.	0.5		0.75	1	0.75	1	1.5	2	2.5	3	1	1.5	2
Brine Tank Size; (inches)	13.5 x 22.5 x 27.25		13.5 x 22.5 x 44	13.5 x 22.5 x 44	16 x 33	16 x 33	16 x 33	16 x 33	16 x 40	16 x 40	16 x 33	16 x 33	16 x 40
Brine Tank Capacity; Lbs NaCl	75		225	225	258	258	258	258	338	338	258	258	338
Drain Line Flow Control; GPM	0.7		1.7	1.7	1.7	2.7	2.7	3.2	5.3	5.3	1.7	2.2	3.2
Brine Line (Re-Fill)Flow Control; GPM	0.5		0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5
Injector; color	1D - Red		1E - White	1E - White	1C - Violet	1E - White	1E - White	1F - Blue	1G - Yellow	1H - Green	1E - White	1G - Yellow	1G - Yellow

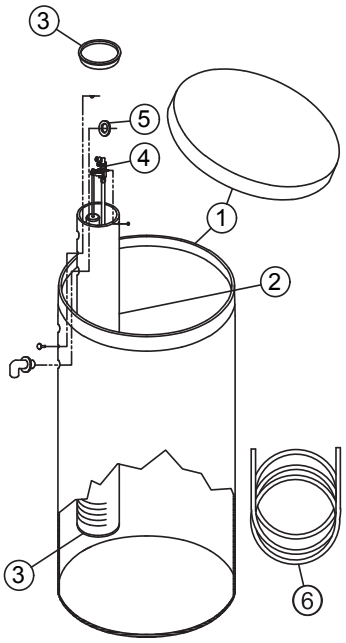
SOFTENER ASSEMBLY



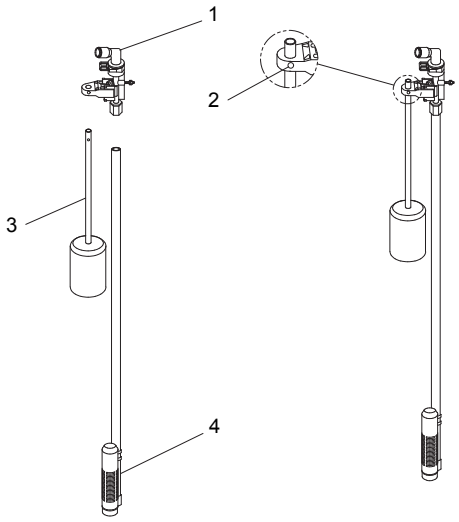
Dwg. No.	Order No.	Description	Qty
1	BPWS1LRV (detailed components shown in this manual)	Brita Custom Softener Valve Complete	1
2	BPD1203	Top Distributor	1
3,4	BPFG844, BP91255	Tank, Pipe, Connector, Assy, for Model BP75J	1
	BPFG1044, BP91255	Tank, Pipe, Connector, Assy for Model BP100J	1
	BPFG1054, BP91255	Tank, Pipe, Connector, Assy for Model BP150J, BPFES100J	1
	BPFG1252, BP91255	Tank, Pipe, Connector, Assy for Models BP200, BPFES150	1
	BPFG1354, BP91255	Tank, Pipe, Connector, Assy for Model BP250, BPFES200	1
	BPFG1465, BP93565	Tank, Pipe, Connector, Assy for Models BP300	1
5,6	BPA410232	Support Bed & Ion Exchange Fine Mesh Resin for BPFES100J	1
	BPA410245	Support Bed & Ion Exchange Fine Mesh Resin for BPFES150	1
	BPA410260	Support Bed & Ion Exchange Fine Mesh Resin for BPFES200	1
	BPA4074	Ion Exchange Resin for BP Models	1

BRINE TANK ASSEMBLY

Dwg. No.	Order No.	Description	Qty
1	BPJ7057	16x33 Assembly for Models BP75J, 100J, 150J, 200, 250; BPFES100J,150, 200	1
	BP71640	16x40 Brine Tank with Cover for Model BP300	1
2	BPH1031	4x30 Slotted Brine Well for Models BP75J, 100J, 150J, 200, 250; BPFES100J,150, 200	1
	BPH1036	4x36 Slotted Brine Well for Model BP300	1
3	BPH7016	4 inch Brine Well Cap for Models BP75J, 100J, 150J, 200, 250; BPFES100J,150, 200	2
		(Top Cap Only for Model BP300)	(1)
4	BP4740	3/8” Brine Valve Assembly (using 1/2”x25” long air check assy for Models BP75J, 100J, 150J, 200, 250; BPFES100J,150, 200)	1
	BP4740L	3/8” Brine Valve Assembly (using 1/2”x43” long air check assy for Model BP300)	1
5	BPH1018	Two Piece Overflow Set	1
6	BPH1023	3/8”O.D.x6 Ft. Poly Tubing	1



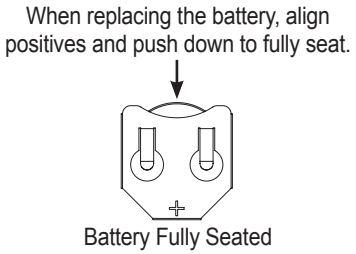
BRINE VALVE ASSEMBLY

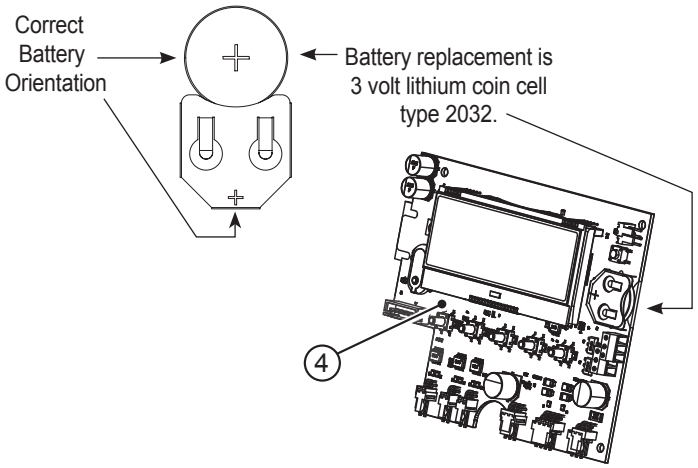


Dwg No.	Order No.	Description	Qty
1	BPH4600	3/8” Safety Brine Valve	1
2	BP10151	Pin	1
3	BPH46409.5	Float Assembly	1
4	BPH450048	Air Check Assembly (uncut length) see brine tank assy for cut length	1

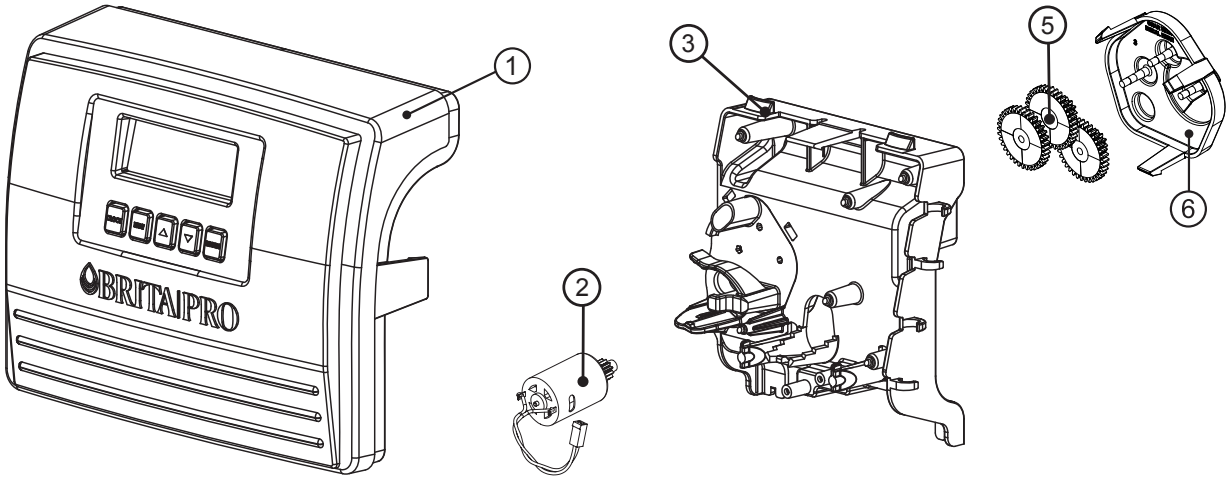
FRONT COVER AND DRIVE ASSEMBLY

Drawing No.	Order No.	Description	Quantity
1	BPVLVCOVER	BRITA PRO VALVE FACE PLATE	
2	BPV310701	MOTOR	1
3	BPV3002-A	DRIVE BRACKET	1
4	BPV4478LR	PC BOARD	1
5	BPV3110	DRIVE GEAR 12X36	3
6	BPV3109	DRIVE GEAR COVER	1





AC Adapter Order No.	(Not Shown) V3186-06
Supply Voltage	100-120 VAC
Supply Frequency	50/60 Hz
Output Voltage	15 VDC
Output Current	500 mA

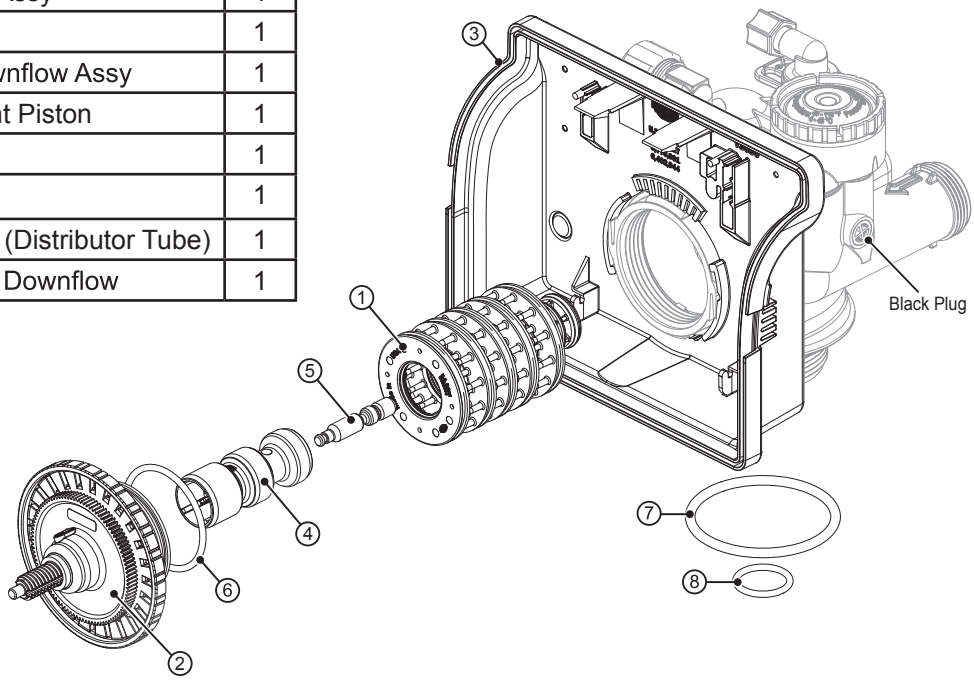


After completing any valve maintenance involving the drive assembly or the drive cap assembly and pistons, unplug power source jack from the printed circuit board (black wire) and plug back in or press and hold NEXT and REGEN buttons for 3 seconds. This resets the electronics and establishes the service piston position. The display should flash the software version and then reset the valve to the service position.

DRIVE CAP ASSEMBLY, DOWNFLOW PISTON, REGENERANT PISTON AND SPACER STACK ASSEMBLY

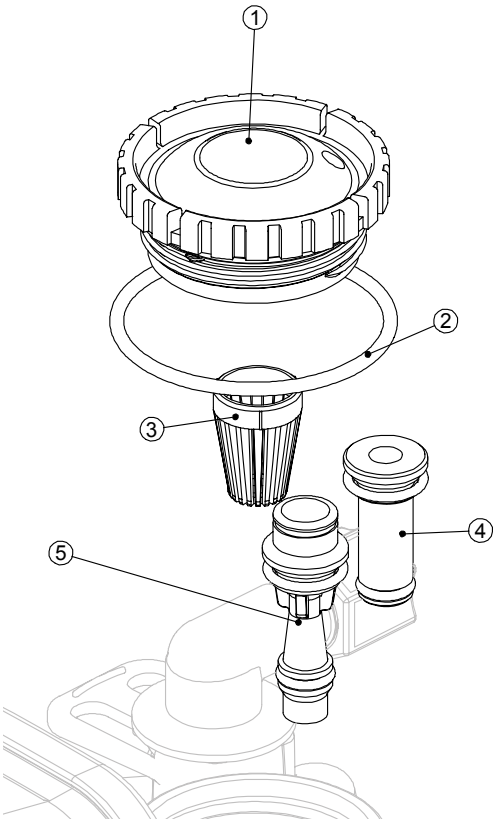
Drawing No.	Order No.	Description	Qty
1	BPV3005	Spacer Stack Assembly	1
2	BPV3004	Drive Cap Assy	1
3	BPV3178LP	Back Plate	1
4	BPV3011	Piston Downflow Assy	1
5	BPV3174	Regenerant Piston	1
6	BPV3135	O-ring 228	1
7	BPV3180	O-ring 337	1
8	BPV3105	O-ring 215 (Distributor Tube)	1
Not Shown	BPV3001	Body Assy Downflow	1

Do not use Vaseline, oils, other hydrocarbon lubricants or spray silicone anywhere. A silicone lubricant may be used on black o-rings. **Avoid any type of lubricants, including silicone, on the clear lip seals.**



After completing any valve maintenance involving the drive assembly or the drive cap assembly and pistons, unplug power source jack from the printed circuit board (black wire) and plug back in or press and hold NEXT and REGEN buttons for 3 seconds. This resets the electronics and establishes the service piston position. The display should flash the software version and then reset the valve to the service position.

INJECTOR CAP, INJECTOR SCREEN, INJECTOR, PLUG AND O-RING



Drawing No.	Order No.	Description	Qty
1	BPV3176	INJECTOR CAP	1
2	BPV3152	O-RING 135	1
3	BPV317701	INJECTOR SCREEN CAGE	1
4	BPV30101Z	INJECTOR ASSY Z PLUG	1
5	BPV30101A	INJECTOR ASSY A BLACK	1
	BPV30101B	INJECTOR ASSY B BROWN	
	BPV30101C	INJECTOR ASSY C VIOLET	
	BPV30101D	INJECTOR ASSY D RED	
	BPV30101E	INJECTOR ASSY E WHITE	
	BPV30101F	INJECTOR ASSY F BLUE	
	BPV30101G	INJECTOR ASSY G YELLOW	
	BPV30101H	INJECTOR ASSY H GREEN	
	BPV30101I	INJECTOR ASSY I ORANGE	
	BPV30101J	INJECTOR ASSY J LIGHT BLUE	
	BPV30101K	INJECTOR ASSY K LIGHT GREEN	
Not Shown	BPV3170	O-RING 011	*
Not Shown	BPV3171	O-RING 013	*

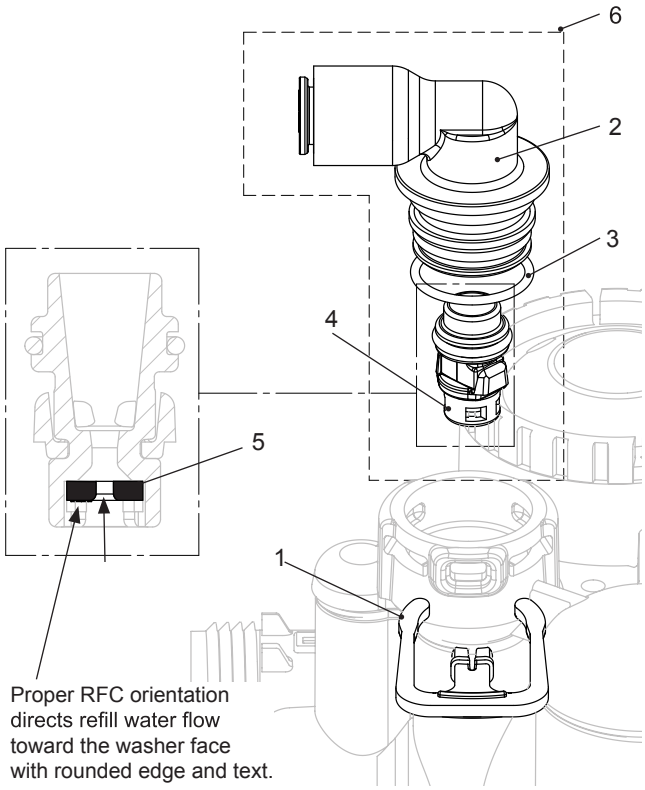
* The injector plug and the injector each contain one 011 (lower) and 013 (upper) o-ring.

The nut and caps are designed to be unscrewed or tightened by hand or with the service spanner wrench (see page 20). If necessary a pliers can be used to unscrew the nut or cap. Do not use a pipe wrench to tighten or loosen nuts or caps. Do not place a screwdriver in slots on caps and/or tap with a hammer.

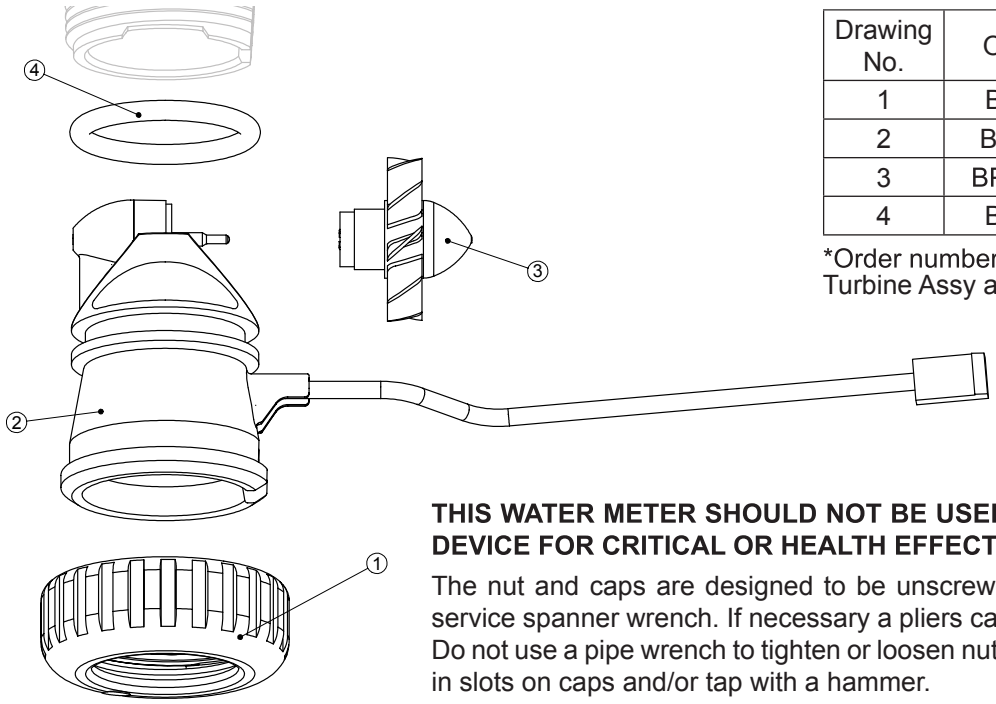
REFILL FLOW CONTROL ASSEMBLY

Drawing No.	Order No.	Description	Qty
1	BPH4615	Elbow Locking Clip	1
2	BPH4628	Elbow 3/8” Liquifit	1
3	BPV3163	O-ring 019	1
4	BPV316501*	RFC Retainer Assy (0.5 gpm)	1
5	BPV3182	RFC	1
6	BPV414401	Elbow 3/8 Liquifit Assy w/RFC	1

*Assembly includes BPV3182 RFC.



WATER METER



Drawing No.	Order No.	Description	Qty
1	BPV3151	Nut 1” QC	1
2	BPV3003*	Meter Assy	1
3	BPV311801	Turbine Assy	1
4	BPV3105	0-ring 215	1

*Order number BPV3003 includes BPV311801 Turbine Assy and BPV3105 O-ring 215.

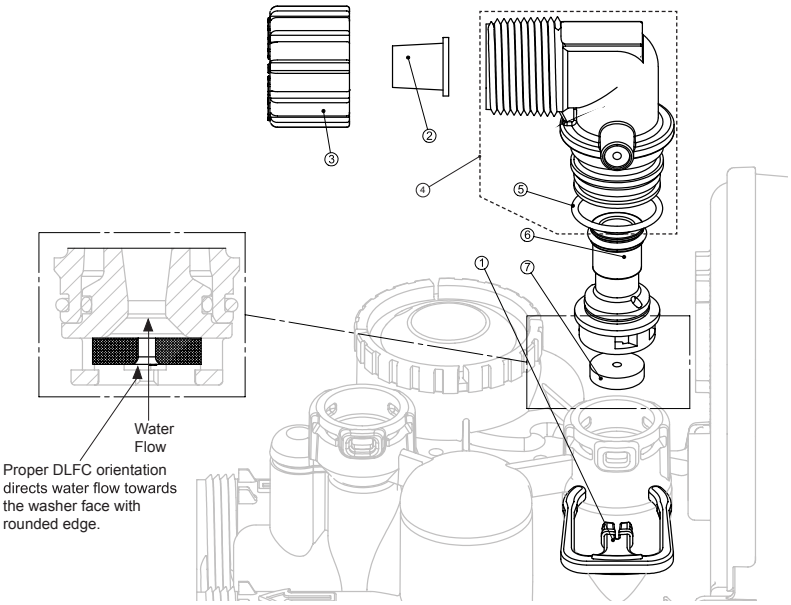
THIS WATER METER SHOULD NOT BE USED AS THE PRIMARY MONITORING DEVICE FOR CRITICAL OR HEALTH EFFECT APPLICATIONS.

The nut and caps are designed to be unscrewed or tightened by hand or with the service spanner wrench. If necessary a pliers can be used to unscrew the nut or cap. Do not use a pipe wrench to tighten or loosen nuts or caps. Do not place a screwdriver in slots on caps and/or tap with a hammer.

DRAIN LINE - 3/4”

Dwg No.	Order No.	Description	Qty
1	BPH4615	Elbow Locking Clip	1
2	BPV3194	Polytube insert 5/8	Option
3	BPV3192	Nut ¾ Drain Elbow	Option
4*	BPV3158	Drain Elbow ¾ Male	1
5	BPV3163	O-ring 019	1
6*	BPV3159	DLFC Retainer Assy	1
7	BPV3162007	DLFC 0.7 gpm for ¾	One DLFC must be used if ¾ fitting is used
	BPV3162010	DLFC 1.0 gpm for ¾	
	BPV3162013	DLFC 1.3 gpm for ¾	
	BPV3162017	DLFC 1.7 gpm for ¾	
	BPV3162022	DLFC 2.2 gpm for ¾	
	BPV3162027	DLFC 2.7 gpm for ¾	
	BPV3162032	DLFC 3.2 gpm for ¾	
	BPV3162042	DLFC 4.2 gpm for ¾	
	BPV3162053	DLFC 5.3 gpm for ¾	
	BPV3162065	DLFC 6.5 gpm for ¾	
	BPV3162075	DLFC 7.5 gpm for ¾	
	BPV3162090	DLFC 9.0 gpm for ¾	
	BPV3162100	DLFC 10.0 gpm for ¾	

*4 and 6 can be ordered as a complete assembly - BPV3331 Drain Elbow and Retainer Assy

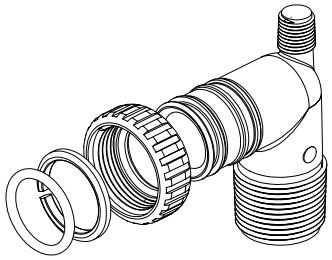


Do not use pipe dope or other sealants on threads. Teflon tape must be used on threads of 3/4” NPT connection (unless using 5/8” polytubing).

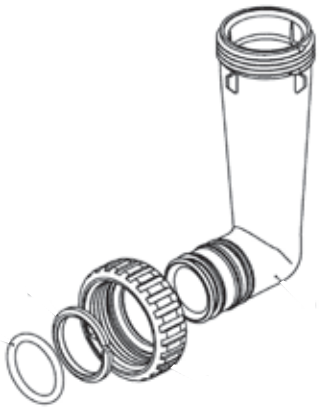
Do not use Vaseline, oils, other hydrocarbon lubricants or spray silicone anywhere. A silicone lubricant may be used on black o-rings.

INSTALLATION FITTING ASSEMBLIES

Order No: **BPV3007** (Standard)
Description: **Fitting 1” PVC Male NPT Elbow Assembly**



Order No. **BPV319101** (Optional)
Description: **Fitting Vertical Adapter Assembly**



Do not use pipe dope or other sealants on threads. Teflon tape must be used on threads of 1” NPT connection. Teflon tape is not necessary on the nut connection nor caps because of o-ring seals.

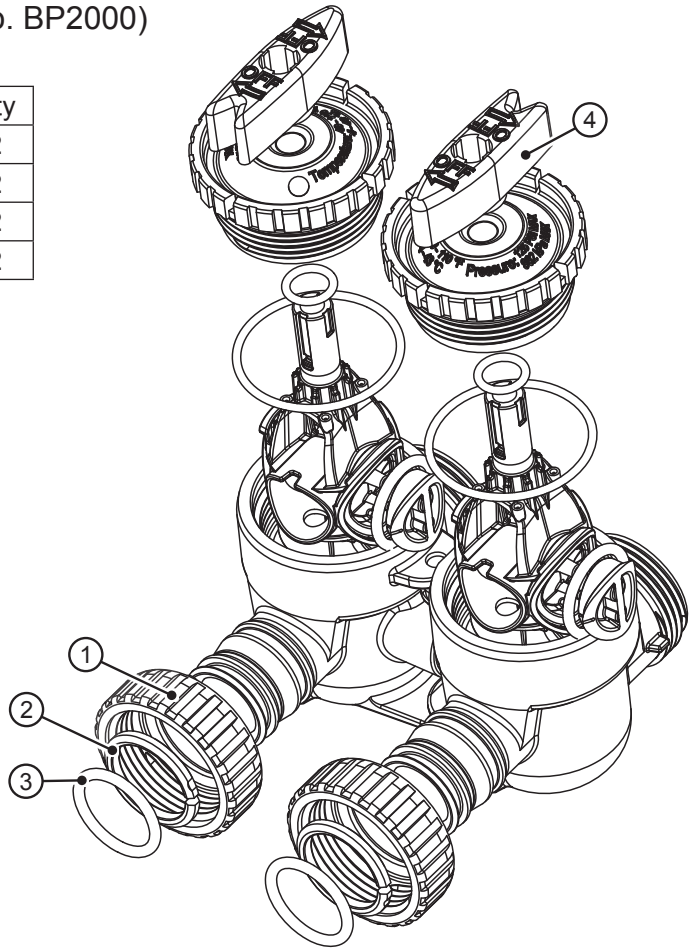
The nut and caps are designed to be unscrewed or tightened by hand or with the service spanner wrench. If necessary a pliers can be used to unscrew the nut or cap. Do not use a pipe wrench to tighten or loosen nuts or caps. Do not place a screwdriver in slots on caps and/or tap with a hammer.

Do not use Vaseline, oils, other hydrocarbon lubricants or spray silicone anywhere. A silicone lubricant may be used on black o-rings.

BYPASS VALVE

(Order No. BP2000)

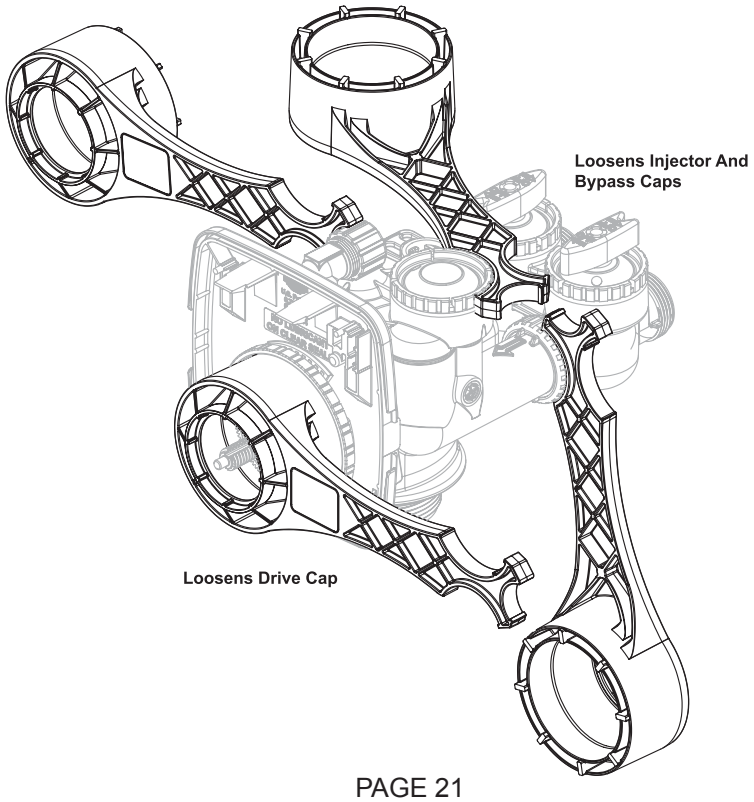
Drawing No.	Order No.	Description	Qty
1	BPV3151	Nut 1” Quick Connect	2
2	BPV3150	Split Ring	2
3	BPV3105	O-Ring 215	2
4	BPV3147	Bypass Handle	2



SERVICE SPANNER WRENCH

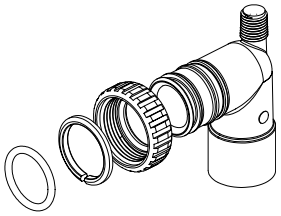
(Order No. BPV3193)

Although no tools are necessary to assemble or disassemble the valve, the wrench (shown in various positions on the valve) may be purchased to aid in assembly or disassembly.

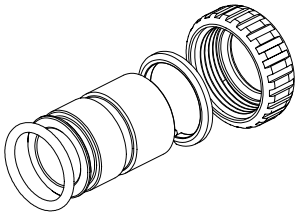


OPTIONAL INSTALLATION FITTING ASSEMBLIES

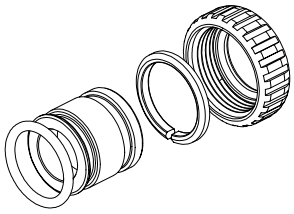
Order No: **BPV300701**
Description: **Fitting ¾" & 1" PVC Solvent 90° Assembly**



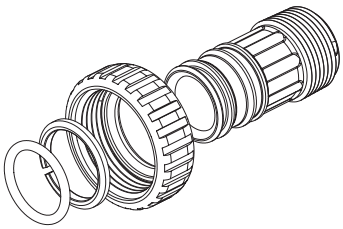
Order No: **BPV300702**
Description: **Fitting 1" Brass Sweat Assembly**
Do not install in California.



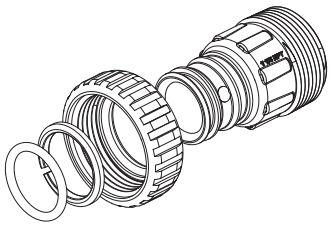
Order No: **BPV300703**
Description: **Fitting ¾" Brass Sweat Assembly**
Do not install in California.



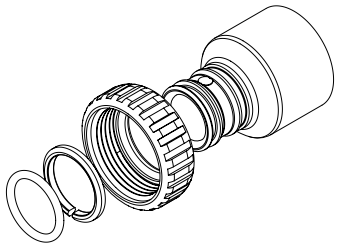
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Description: **Fitting 1" Plastic Male NPT Assembly**



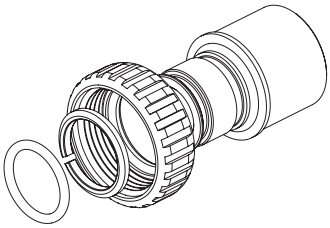
Order No: **BPV300705**
Description: **Fitting 1-¼" Plastic Male NPT Assembly**



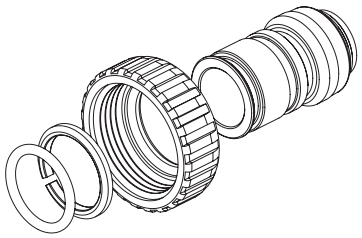
Order No. **BPV300707**
Description: **Fitting 1¼" & 1½" PVC Solvent Assembly**



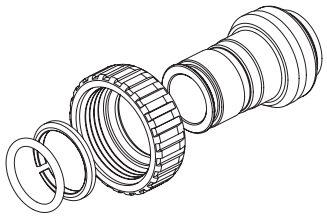
Order No: **BPV300709**
Description: **Fitting 1¼" & 1½" Brass Sweat Assembly**



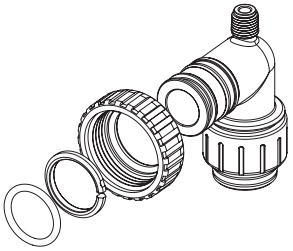
Order No. **BPV300712**
Description: **Fitting 3/4" Brass SharkBite Assembly**



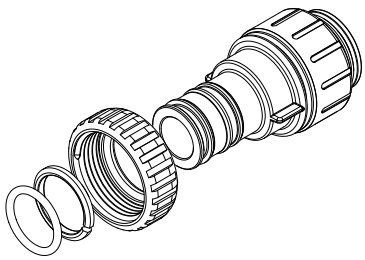
Order No. **BPV300713**
Description: **Fitting 1» Brass SharkBite Assembly**



Order No: **BPV300715**
Description: **WS1 FTG 3/4 JG QC 90 Assembly**



Order No: **BPV300717**
Description: **WS1 FTG 1" JG QC Assembly**



SERVICE INSTRUCTIONS



When servicing the valve, water may leak from the valve. Water from the valve may create a slip hazard. Clean up water spills.



Disconnect from electrical power prior to servicing the valve.

DRIVE ASSEMBLY:

Remove the valve cover to access the drive assembly.

Disconnect the power source plug (black wire) from the PC board prior to disconnecting the motor or water meter plugs from the PC board. The power source plug connects to the four-pin jack. The motor plug connects to the two-pin jack on the left-hand side of the PC board. The water meter plug (gray wire) connects to the three-pin jack on the far right-hand side of the PC board.

The PC board can be removed separately from the drive bracket but it is not recommended. Do not attempt to remove the display panel from the PC board. Handle the board by the edges. To remove the PC board from the drive bracket, unplug the power, water meter and motor plugs from the PC board. Lift the middle latch along the top of the drive bracket while pulling outward on the top of the PC board. The drive bracket has two plastic pins that fit into the holes on the lower edge of the PC board. Once the PC board is tilted about 45° from the drive bracket it can be lifted off of these pins. To reinstall the PC board, position the lower edge of the PC board so that the holes in the PC board line up with the plastic pins. Push the top of the PC board towards the valve until it snaps under the middle latch, weave the power and water meter wires into the holders and reconnect the motor, water meter and power plugs.

The drive bracket must be removed to access the drive cap assembly and pistons or the drive gear cover. It is not necessary to remove the PC board from the drive bracket to remove the drive bracket. To remove the drive bracket start by removing the plugs for the power source and the water meter. Unweave the wires from the side holders. Two tabs on the top of the drive back plate hold the drive bracket in place. Simultaneously lift the two tabs and gently ease the top of the drive bracket forward. The lower edge of the drive bracket has two notches that rest on the drive back plate. Lift up and outward on the drive bracket to disengage the notches.

To reassemble, seat the bottom of the drive bracket so the notches are engaged at the bottom of the drive back plate. Push the top of the drive bracket toward the two latches. The drive bracket may have to be lifted slightly to let the threaded piston rod pass through the hole in the drive bracket. Maintain a slight engaging force on top of the drive bracket while deflecting the bracket slightly to the left by pressing on the side of the upper right corner. This helps the drive gears mesh with the drive cap assembly. The drive bracket is properly seated when it snaps under the latches on the drive back plate. If resistance is felt before latching, then notches are not fully engaged, the piston rod is not in hole, the wires are jammed between the drive bracket and drive back plate, or the gear is not engaging the drive cap assembly.

To inspect the drive gears, the drive gear cover needs to be removed. Before trying to remove the gear cover, the drive bracket must be removed from the drive back plate. (Refer to the instructions above regarding removing the drive bracket from the drive back plate. The drive gear cover can be removed from the drive bracket without removing the motor or the PC board.) The drive gear cover is held in place on the drive bracket by three clips. The largest of the three clips is always orientated to the bottom of the drive bracket. With the PC board facing up, push in and down on the large clip on the drive gear cover. Handle the cover and the gears carefully so that the gears do not fall off the pegs in the cover.

Replace broken or damaged drive gears. Do not lubricate any of the gears. Avoid getting any foreign matter on the reflective coating because dirt or oils may interfere with pulse counting.

The drive gear cover only fits on one way, with the large clip orientated towards the bottom. If all three clips are outside of the gear shroud on the drive bracket the drive gear cover slips easily into place.

The drive bracket does not need to be removed from the drive plate if the motor needs to be removed. To remove the motor, disconnect the power and motor plugs from the jacks on the PC board. Pull the locking lever away from the motor and pull the motor away from the locking tab then straight out of the drive bracket.

Replace the motor if necessary. Do not lubricate the motor or the gears. To reinstall the motor, place the motor gear side down into the end of the slot in the drive bracket furthest away from the locking tab. Ensure the holes in the motor housing are a quarter-turn away from the locking tab. Ensure the motor is flush with the drive bracket before proceeding. Gently rotate the motor clockwise (to the right) while pushing the motor towards the locking tab. The motor will engage the gears in the drive bracket while sliding into place. Ensure the locking tab on the drive bracket is fully inserted into one of the holes in the motor housing. Reconnect the power plug.

Replace the valve cover. After completing any valve maintenance involving the drive assembly or the drive cap assembly and pistons unplug power source jack from the printed circuit board (black wire) and plug back in or press and hold NEXT and REGEN buttons for 3 seconds.

This resets the electronics and establishes the service piston position. The display should flash all wording, then flash the software version and then reset the valve to the service position.

DRIVE CAP ASSEMBLY, MAIN PISTON AND REGENERANT PISTON:

The drive assembly must be removed to access the drive cap assembly. The drive cap assembly must be removed to access the piston(s). The drive cap assembly is threaded into the control valve body and seals with an o-ring. To remove the drive cap assembly use the special plastic wrench or insert a ¼" to ½" flat blade screwdriver into one of the slots around the top 2" of the drive cap assembly so it engages the notches molded into the drive back plate around the top 2" of the piston cavity. See below. The notches are visible through the holes. Lever the screwdriver so the drive cap assembly turns counter clockwise. Once loosened unscrew the drive cap assembly by hand and pull straight out.

The drive cap assembly contains the drive cap, the main drive gear, drive cap spline, piston rod and various other parts that should not be dissembled in the field. The only replaceable part on the drive cap assembly is the o-ring. Attached to the drive cap assembly is the main piston and if a regenerant is used, a regenerant piston.

The regenerant piston (the small diameter one behind the main piston) is removed from the main piston by pressing sideways and unsnapping it from its latch. Chemically clean in dilute sodium bisulfite or vinegar, or replace the regenerant piston if needed. To remove the main piston fully extend the piston rod and then unsnap the main piston from its latch by pressing on the side with the number. Chemically clean in dilute sodium bisulfite or vinegar, or replace the main piston.

Reattach the main piston to the drive cap assembly. Reattach the regenerant piston (if needed) to the main piston. Reinsert the drive cap assembly and piston into the spacer stack assembly and hand tighten the drive cap assembly. Continue to tighten the drive cap assembly using a screwdriver as a ratchet until the black o-ring on the spacer stack assembly is no longer visible through the drain port. Excessive force can break the notches molded into the drive back plate. Make certain that the main drive gear still turns freely.

The exact position of the piston is not important as long as the main drive gear turns freely. Reattach the drive assembly to the control valve and connect all plugs. After completing any valve maintenance involving the drive assembly or the drive cap assembly and pistons unplug power source jack from the printed circuit board (black wire) and plug back in or press and hold NEXT and REGEN buttons for 3 seconds.
This resets the electronics and establishes the service piston position. The display should flash all wording, then flash the software version and then reset the valve to the service position.

REFILL FLOW CONTROL ASSEMBLY:

To clean or replace the refill flow control, pull out the elbow-locking clip and then pull straight up on the elbow. Replace the elbow locking clip in the slot so that it is not misplaced. Twist to remove the white flow control retainer. The flow control can be removed by prying upward through the side slots of the retainer with a small flat blade screwdriver.

Chemically clean the flow control or the white flow control retainer using dilute sodium bisulfite or vinegar. Do not use a wire brush. If necessary, replace the flow control, o-ring on the flow control retainer, or the o-ring on the elbow.

Reseat the flow control so the rounded end is visible in the flow control. Reseat the white flow control retainer by pushing the retainer into the elbow until the o-ring seats. Remove locking clip, push down on elbow to reseat and insert locking clip.

Do not use Vaseline, oils, or other unacceptable lubricants on o-rings. A silicone lubricant may be used on the o-ring on the elbow or the white retainer.

SPACER STACK ASSEMBLY:

To access the spacer stack assembly remove the drive assembly, drive cap assembly and piston. The spacer stack assembly can be removed easily without tools by using thumb and forefinger. Inspect the black o-rings and clear lip seals for wear or damage. Replace the entire stack if necessary. Do not disassemble the stack.

The spacer stack assembly may be chemically cleaned (dilute sodium bisulfite or vinegar) or wiped with a soft cloth.

The spacer stack assembly can be pushed in to the control valve body bore by hand. Since the spacer stack assembly can be compressed it is easier to use a blunt object (5/8" to 1-1/8" in diameter) to push the center of the assembly into the control valve body. The assembly is properly seated when at least four threads are exposed (approximately 5/8"). Do not force the spacer stack assembly in. The control valve body bore interior can be lubricated with silicone to allow for easy insertion of the entire stack.

Reattach the drive cap assembly and piston(s) and the drive assembly.

After completing any valve maintenance involving the drive assembly or the drive cap assembly and pistons unplug power source jack from the printed circuit board (black wire) and plug back in or press and hold NEXT and REGEN buttons for 3 seconds.
This resets the electronics and establishes the service piston position. The display should flash all wording, then flash the software version and then reset the valve to the service position.

INJECTOR CAP, SCREEN, INJECTOR PLUG AND INJECTOR:

Unscrew the injector cap and lift off. Loosen cap with special plastic wrench or pliers if necessary. Attached to the injector cap is a screen. Remove the screen and clean if fouled.

The plug and/or injector can be pried out with a small screwdriver. The plug can be wiped clean. If the plug leaks replace the entire plug. The injector consists of a throat and a nozzle. Chemically clean the injector with vinegar or sodium bisulfite. The holes can be blown out with air. Both pieces have small diameter holes that control the flow rates of water to insure that the proper concentration of regenerant is used. Sharp objects, which can score the plastic, should not be used to clean the injector. Scoring the injector or increasing the diameter of the hole could change the operating parameters of the injector.

Push the plug(s) and/or injectors firmly in place, replace the screen and hand tighten the injector cap.

WATER METER:

The water meter assembly is connected to the PC board by a wire. If the entire water meter assembly is to be replaced, remove the control valve cover and disconnect the power source and water meter plugs from the PC board. Unlatch the drive assembly and lean it forward. Unthread the water meter wire from the side of the drive assembly and through the drive back plate. To reinstall, rethread the water meter wire through the drive back plate and the side of the drive assembly. Reattach the drive assembly and the water meter and power plugs.

THIS WATER METER SHOULD NOT BE USED AS THE PRIMARY MONITORING DEVICE FOR CRITICAL OR HEALTH EFFECT APPLICATIONS.

OPERATING PRESSURES:
30 PSI MINIMUM / 80 PSI MAXIMUM

OPERATING TEMPERATURES:
40° F MINIMUM / 110° F MAXIMUM

If no water meter wire is visible, then a plug is installed, not a water meter.

The water meter wire does not need to be removed from the PC board if the water meter is only being inspected and cleaned. To remove the water meter assembly, unscrew the meter cap on the left side of the control valve. Pliers may be used to unscrew the nut if necessary.

With the nut removed, a slot at the top of the water meter is visible. Twist a flat blade screwdriver in the slot between the control valve body and the meter. When the meter is part way out it is easy to remove the water meter from the housing. Once the water meter is removed from the control valve body, gently pull forward on the turbine to remove it from the shaft. Do not use a wire brush to clean the turbine. Wipe with a clean cloth or chemically clean in dilute sodium bisulfite or vinegar. The turbine can be immersed in the chemical. Do not immerse electronics. If the turbine is scored or damaged or the bearings on the turbine are worn, replace the turbine.

Do not lubricate the turbine shaft. The turbine shaft bearings are prelubricated. Do not use Vaseline, oils, or other unacceptable lubricants on the o-ring. A silicone lubricant may be used on the black o-ring.

Snap the turbine on the shaft and reinsert the water meter into the side slot. Hand tighten the nut. Do not use a pipe wrench to tighten nut.

TROUBLESHOOTING

Problem	Possible Cause	Solution
1. No Display on PC Board	a. No power at electric outlet	a. Repair outlet or use working outlet
	b. Control valve Power Adapter not plugged into outlet or power cord end not connected to PC board connection	b. Plug Power Adapter into outlet or connect power cord end to PC Board connection
	c. Improper power supply	c. Verify proper voltage is being delivered to PC Board
	d. Defective Power Adapter	d. Replace Power Adapter
	e. Defective PC Board	e. Replace PC Board
2. PC Board does not display correct time of day	a. Power Adapter plugged into electric outlet controlled by light switch	a. Use uninterrupted outlet
	b. Tripped breaker switch and/or tripped GFI	b. Reset breaker switch and/ or GFI switch
	c. Power outage	c. Reset time of day. If PC Board has battery back up present the battery may be depleted. See Front Cover and Drive Assembly drawing for instructions.
	d. Defective PC Board	d. Replace PC Board
3. Display does not indicate that water is flowing. Refer to user instructions for how the display indicates water is flowing	a. Bypass valve in bypass position	a. Turn bypass handles to place bypass in service position
	b. Meter is not connected to meter connection on PC Board	b. Connect meter to three pin connection labeled METER on PC Board
	c. Restricted/ stalled meter turbine	c. Remove meter and check for rotation or foreign material
	d. Meter wire not installed securely into three pin connector	d. Verify meter cable wires are installed securely into three pin connector labeled METER
	e. Defective meter	e. Replace meter
	f. Defective PC Board	f. Replace PC Board
4. Control valve regenerates at wrong time of day	a. Power outage	a. Reset time of day. If PC Board has battery back up present the battery may be depleted. See Front Cover and Drive Assembly drawing for instructions.
	b. Time of day not set correctly	b. Reset to correct time of day
	c. Time of regeneration set incorrectly	c. Reset regeneration time
	d. Control valve set at “on 0” (immediate regeneration)	d. Check programming setting and reset to NORMAL (for a delayed regen time)
	e. Control valve set at “NORMAL + on 0” (delayed and/ or immediate)	e. Check programming setting and reset to NORMAL (for a delayed regen time)
5. Time of day flashes on and off	a. Power outage	a. Reset time of day. If PC Board has battery back up present the battery may be depleted. See Front Cover and Drive Assembly drawing for instructions.
6. Control valve does not regenerate automatically when the REGEN button	a. Broken drive gear or drive cap assembly	a. Replace drive gear or drive cap assembly
	b. Broken Piston Rod	b. Replace piston rod
	c. Defective PC Board	c. Defective PC Board
7. Control valve does not regenerate automatically but **does** when the REGEN button	a. Bypass valve in bypass position	a. Turn bypass handles to place bypass in service position
	b. Meter is not connected to meter connection on PC Board	b. Connect meter to three pin connection labeled METER on PC Board
	c. Restricted/ stalled meter turbine	c. Remove meter and check for rotation or foreign material
	d. Incorrect programming	d. Check for programming error
	e. Meter wire not installed securely into three pin connector	e. Verify meter cable wires are installed securely into three pin connector labeled METER
	f. Defective meter	f. Replace meter
	g. Defective PC Board	g. Replace PC Board

Problem	Possible Cause	Solution
8. Hard or untreated water is being delivered	a. Bypass valve is open or faulty	a. Fully close bypass valve or replace
	b. Media is exhausted due to high water usage	b. Check program settings or diagnostics for abnormal water usage
	c. Meter not registering	c. Remove meter and check for rotation or foreign material
	d. Water quality fluctuation	d. Test water and adjust program values accordingly
	e. No regenerant or low level of regenerant in regenerant tank	e. Add proper regenerant to tank
	f. Control fails to draw in regenerant	f. Refer to Trouble Shooting Guide number 12
	g. Insufficient regenerant level in regenerant tank	g. Check refill setting in programming. Check refill flow control for restrictions or debris and clean or replace
	h. Damaged seal/stack assembly	h. Replace seal/stack assembly
	i. Control valve body type and piston type mix matched	i. Verify proper control valve body type and piston type match
	j. Fouled media bed	j. Replace media bed
9. Control valve uses too much regenerant	a. Improper refill setting	a. Check refill setting
	b. Improper program settings	b. Check program setting to make sure they are specific to the water quality and application needs
	c. Control valve regenerates frequently	c. Check for leaking fixtures that may be exhausting capacity or system is undersized
10. Residual regenerant being delivered to service	a. Low water pressure	a. Check incoming water pressure – water pressure must remain at minimum of 25 psi
	b. Incorrect injector size	b. Replace injector with correct size for the application
	c. Restricted drain line	c. Check drain line for restrictions or debris and clean
11. Excessive water in regenerant tank	a. Improper program settings	a. Check refill setting
	b. Plugged injector	b. Remove injector and clean or replace
	c. Drive cap assembly not tightened in properly	c. Re-tighten the drive cap assembly
	d. Damaged seal/ stack assembly	d. Replace seal/ stack
	e. Restricted or kinked drain line	e. Check drain line for restrictions or debris and or un-kink drain line
	f. Plugged backwash flow controller	f. Remove backwash flow controller and clean or replace
	g. Missing refill flow controller	g. Replace refill flow controller
12. Control valve fails to draw in regenerant	a. Injector is plugged	a. Remove injector and clean or replace
	b. Faulty regenerant piston	b. Replace regenerant piston
	c. Regenerant line connection leak	c. Inspect regenerant line for air leak
	d. Drain line restriction or debris cause excess back pressure	d. Inspect drain line and clean to correct restriction
	e. Drain line too long or too high	e. Shorten length and or height
	f. Low water pressure	f. Check incoming water pressure – water pressure must remain at minimum of 25 psi

Problem	Possible Cause	Solution
13. Water running to drain	a. Power outage during regeneration	a. Upon power being restored control will finish the remaining regeneration time. Reset time of day.
	b. Damaged seal/ stack assembly	b. Replace seal/ stack assembly
	c. Piston assembly failure	c. Replace piston assembly
	d. Drive cap assembly not tightened in properly	d. Re-tighten the drive cap assembly
14. Err – 101 = Control unable to sense motor movement	a. Motor not inserted full to engage pinion, motor wires broken or disconnected	a. Disconnect power, make sure motor is fully engaged, check for broken wires, make sure two pin connector on motor is connected to the two pin connection on the PC Board labeled MOTOR. Press NEXT and REGEN buttons for 3 seconds to resynchronize software with piston position or disconnect power supply from PC Board for 5 seconds and then reconnect.
	b. PC Board not properly snapped into drive bracket	b. Properly snap PC Board into drive bracket and then Press NEXT and REGEN buttons for 3 seconds to resynchronize software with piston position or disconnect power supply from PC Board for 5 seconds and then reconnect.
	c. Missing reduction gears	c. Replace missing gears
15. Err – 102 = Control valve motor ran too short and was unable to find the next cycle position and stalled	a. Foreign material is lodged in control valve	a. Open up control valve and pull out piston assembly and seal/ stack assembly for inspection. Press NEXT and REGEN buttons for 3 seconds to resynchronize software with piston position or disconnect power supply from PC Board for 5 seconds and then reconnect.
	b. Mechanical binding	b. Check piston and seal/ stack assembly, check reduction gears, check drive bracket and main drive gear interface. Press NEXT and REGEN buttons for 3 seconds to resynchronize software with piston position or disconnect power supply from PC Board for 5 seconds and then reconnect.
	c. Main drive gear too tight	c. Loosen main drive gear. Press NEXT and REGEN buttons for 3 seconds to resynchronize software with piston position or disconnect power supply from PC Board for 5 seconds and then reconnect.
	d. Improper voltage being delivered to PC Board	d. Verify that proper voltage is being supplied. Press NEXT and REGEN buttons for 3 seconds to resynchronize software with piston position or disconnect power supply from PC Board for 5 seconds and then reconnect.

Problem	Possible Cause	Solution
16. Err – 103 = Control valve motor ran too long and was unable to find the next cycle position	a. Motor failure during a regeneration	a. Check motor connections then Press NEXT and REGEN buttons for 3 seconds to resynchronize software with piston position or disconnect power supply from PC Board for 5 seconds and then reconnect.
	b. Foreign matter built up on piston and stack assemblies creating friction and drag enough to time out motor	b. Replace piston and stack assemblies. Press NEXT and REGEN buttons for 3 seconds to resynchronize software with piston position or disconnect power supply from PC Board for 5 seconds and then reconnect.
	c. Drive bracket not snapped in properly and out enough that reduction gears and drive gear do not interface	c. Snap drive bracket in properly then Press NEXT and REGEN buttons for 3 seconds to resynchronize software with piston position or disconnect power supply from PC Board for 5 seconds and then reconnect.
17. Err – 104 = Control valve motor ran too long and timed out trying to reach home position	a. Drive bracket not snapped in properly and out enough that reduction gears and drive gear do not interface	a. Snap drive bracket in properly then Press NEXT and REGEN buttons for 3 seconds to resynchronize software with piston position or disconnect power supply from PC Board for 5 seconds and then reconnect.



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www.britapro.com

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